

Babu Banarasi Das University



NO SQL PROJECT ON

Student Record Management System

Submitted to: Mr. Harendra Singh

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OBJECTIVES:

- To understand the concept of NoSQL databases.
- To create a database and collection in MongoDB.
- To insert and manage 10 student records.
- To perform CRUD operations — Create, Read, Update, and Delete.
- To use comparison operators such as \$gt, \$gte, \$lt, \$lte, \$ne, and \$in.
- To perform sorting and filtering of student records.
- To perform search and retrieval operations based on specific conditions.
- To understand and implement AND (\$and) and OR (\$or) operations.
- To understand how MongoDB can be used to efficiently manage and retrieve student information.

1. CREATE DATABASE

MongoDB Query:

> *use AcademicDB*

Connections Edit View Help

Compass

{ } My Queries

Data Modeling

> _ mongosh: AcademicDB

> _MONGOSH

> use AcademicDB

< switched to db AcademicDB

2. CREATE COLLECTION

MongoDB Query:

> *db.createCollection("students")*

```
> db.createCollection("students")
< { ok: 1 }
```

3. INSERT 10 STUDENTS

MongoDB Query:

> *db.students.insertMany([])*

```
db.students.insertMany([
  {roll:1, name:"Akash Shukla", age:19, marks:85, city:"Lucknow"},
  {roll:2, name:"Priya Singh", age:18, marks:78, city:"Delhi"},
  {roll:3, name:"Rohan Gupta", age:19, marks:91, city:"Kanpur"},
  {roll:4, name:"Ishita Verma", age:18, marks:66, city:"Delhi"},
  {roll:5, name:"Manish Tiwari", age:19, marks:82, city:"Lucknow"},
  {roll:6, name:"Divya Pandey", age:20, marks:88, city:"Kanpur"},
  {roll:7, name:"Saurabh Dubey", age:19, marks:71, city:"Lucknow"},
  {roll:8, name:"Neelam Rathore", age:20, marks:93, city:"Delhi"},
  {roll:9, name:"Kunal Saxena", age:18, marks:54, city:"Kanpur"},
  {roll:10, name:"Tanvi Agarwal", age:20, marks:59, city:"Delhi"}
])
```

CONNECTIONS (1)

▼ Akash

- ▶ admin
- ▶ config
- ▶ local

```
> db.students.insertMany([
  {roll:1, name:"Akash Shukla", age:19, marks:85, city:"Lucknow"},
  {roll:2, name:"Priya Singh", age:18, marks:78, city:"Delhi"},
  {roll:3, name:"Rohan Gupta", age:19, marks:91, city:"Kanpur"},
  {roll:4, name:"Ishita Verma", age:18, marks:66, city:"Delhi"},
  {roll:5, name:"Manish Tiwari", age:19, marks:82, city:"Lucknow"},
  {roll:6, name:"Divya Pandey", age:20, marks:88, city:"Kanpur"},
  {roll:7, name:"Saurabh Dubey", age:19, marks:71, city:"Lucknow"},
  {roll:8, name:"Neelam Rathore", age:20, marks:93, city:"Delhi"},
  {roll:9, name:"Kunal Saxena", age:18, marks:54, city:"Kanpur"},
  {roll:10, name:"Tanvi Agarwal", age:20, marks:59, city:"Delhi"}
])
```

4. DISPLAY ALL STUDENTS

MongoDB Query:

➤ `db.students.find()`

```
> db.students.find()
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e000'),
  roll: 1,
  name: 'Akash Shukla',
  age: 19,
  marks: 85,
  city: 'Lucknow'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e001'),
  roll: 2,
  name: 'Priya Singh',
  age: 18,
  marks: 78,
  city: 'Delhi'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e002'),
  roll: 3,
  name: 'Rohan Gupta',
  age: 19,
  marks: 91,
  city: 'Kanpur'
}
```

5. EQUAL TO — \$eq

MongoDB Query:

> `db.students.find({marks: {$eq: 78}})`

```
> db.students.find({marks: {$eq: 78}})
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e001'),
  roll: 2,
  name: 'Priya Singh',
  age: 18,
  marks: 78,
  city: 'Delhi'
}
```

6. GREATER THAN — \$gt

MongoDB Query:

> `db.students.find({marks: {$gt: 60}})`

Explanation: Finds students whose marks are greater than 60.

```
> db.students.find({marks: {$gt: 60}})
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e000'),
  roll: 1,
  name: 'Akash Shukla',
  age: 19,
  marks: 85,
  city: 'Lucknow'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e001'),
  roll: 2,
  name: 'Priya Singh',
  age: 18,
  marks: 78,
  city: 'Delhi'
}
```

7. LESS THAN — \$lt

MongoDB Query:

> `db.students.find({marks: {$lt: 60}})`

```
> db.students.find({marks: {$lt: 60}})
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e008'),
  roll: 9,
  name: 'Kunal Saxena',
  age: 18,
  marks: 54,
  city: 'Kanpur'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e009'),
  roll: 10,
  name: 'Tanvi Agarwal',
  age: 20,
  marks: 59,
  city: 'Delhi'
}
AcademicDB >
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

➤ `db.students.find({marks: {$gte: 90}})`

```
> db.students.find({marks: {$gte: 90}})
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e002'),
  roll: 3,
  name: 'Rohan Gupta',
  age: 19,
  marks: 91,
  city: 'Kanpur'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e007'),
  roll: 8,
  name: 'Neelam Rathore',
  age: 20,
  marks: 93,
  city: 'Delhi'
}
```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

➤ `db.students.find({marks: {$lte: 70}})`

```
> db.students.find({marks: {$lte: 70}})
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e003'),
  roll: 4,
  name: 'Ishita Verma',
  age: 18,
  marks: 66,
  city: 'Delhi'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e008'),
  roll: 9,
  name: 'Kunal Saxena',
  age: 18,
  marks: 54,
  city: 'Kanpur'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e009'),
  roll: 10,
  name: 'Tanvi Agarwal',
  age: 20,
  marks: 59,
  city: 'Delhi'
}
```

10. UPDATE OPERATION

MongoDB Query:

```
> db.students.updateOne(  
  {roll:4},  
  {$set:{marks:72}}  
)
```

```
> db.students.updateOne({roll: 4}, {$set: {marks: 72}})  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
AcademicDB >
```

```
> db.students.find({roll: 4})  
< {  
  _id: ObjectId('6b3a7f20d1c94e8b5a60e003'),  
  roll: 4,  
  name: 'Ishita Verma',  
  age: 18,  
  marks: 72,  
  city: 'Delhi'  
}  
AcademicDB >
```

11. DELETE ONE OPERATION

MongoDB Query:

```
> db.students.deleteOne({roll:10})
```

```
> db.students.deleteOne({roll: 10})  
< {  
  acknowledged: true,  
  deletedCount: 1  
}  
AcademicDB >
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({  
  marks: {$lt:60}  
})
```

```
> db.students.deleteMany({marks: {$lt: 60}})  
< {  
  acknowledged: true,  
  deletedCount: 1  
}  
AcademicDB >
```

13. AND OPERATION — \$and

MongoDB Query:

```
> db.students.find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
```

```
> db.students.find({ $and: [ {marks: {$gt: 80}}, {age: 19} ] })
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e000'),
  roll: 1,
  name: 'Akash Shukla',
  age: 19,
  marks: 85,
  city: 'Lucknow'
}
```

14. OR OPERATION — \$or

MongoDB Query:

```
> db.students.find({
  $or:[
    {city:"Lucknow"},
    {city:"Kanpur"} ]})
```

```
> db.students.find({ $or: [ {city: "Lucknow"}, {city: "Kanpur"} ] })
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e000'),
  roll: 1,
  name: 'Akash Shukla',
  age: 19,
  marks: 85,
  city: 'Lucknow'
}
```

15. SORT OPERATION

MongoDB Ascending order Query:

➤ `db.students.find().sort({marks : 1})`

```
> db.students.find().sort({marks: 1})
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e006'),
  roll: 7,
  name: 'Saurabh Dubey',
  age: 19,
  marks: 71,
  city: 'Lucknow'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e003'),
  roll: 4,
  name: 'Ishita Verma',
  age: 18,
  marks: 72,
  city: 'Delhi'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e001'),
  roll: 2,
  name: 'Priya Singh',
  age: 18,
  marks: 78,
  city: 'Delhi'
}
```

B. MongoDB Descending Order Query:

> `db.students.find().sort({marks: -1})`

```
> db.students.find().sort({marks: -1})
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e007'),
  roll: 8,
  name: 'Neelam Rathore',
  age: 20,
  marks: 93,
  city: 'Delhi'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e002'),
  roll: 3,
  name: 'Rohan Gupta',
  age: 19,
  marks: 91,
  city: 'Kanpur'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e005'),
  roll: 6,
  name: 'Divya Pandey',
  age: 20,
  marks: 88,
  city: 'Kanpur'
}
```

16. LIMIT OPERATION

MongoDB Query:

➤ `db.students.find().limit(5)`

```
> db.students.find().limit(5)
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e000'),
  roll: 1,
  name: 'Akash Shukla',
  age: 19,
  marks: 85,
  city: 'Lucknow'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e001'),
  roll: 2,
  name: 'Priya Singh',
  age: 18,
  marks: 78,
  city: 'Delhi'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e002'),
  roll: 3,
  name: 'Rohan Gupta',
  age: 19,
  marks: 91,
  city: 'Kanpur'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e003'),
  roll: 4,
  name: 'Ishita Verma',
  age: 18,
  marks: 72,
  city: 'Delhi'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e004'),
  roll: 5,
  name: 'Manish Tiwari',
  age: 19,
  marks: 82,
  city: 'Lucknow'
}
```

17. SEARCH BY CITY

MongoDB Query:

➤ `db.students.find({city:"Lucknow"})`

```
> db.students.find({city:"Lucknow"})
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e000'),
  roll: 1,
  name: 'Akash Shukla',
  age: 19,
  marks: 85,
  city: 'Lucknow'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e004'),
  roll: 5,
  name: 'Manish Tiwari',
  age: 19,
  marks: 82,
  city: 'Lucknow'
}
{
  _id: ObjectId('6b3a7f20d1c94e8b5a60e006'),
  roll: 7,
  name: 'Saurabh Dubey',
  age: 19,
  marks: 71,
  city: 'Lucknow'
}
```

18. COUNT DOCUMENTS

MongoDB Query:

➤ `db.students.countDocuments()`

```
> db.students.countDocuments()
< 8
```

19. UPDATE MANY OPERATION

MongoDB Query:

> *db.students.updateMany({city: "Lucknow"}, {\$set: {state: "UP"}})*

```
> db.students.updateMany({city: "Lucknow"}, {$set: {state: "UP"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

```
> db.students.findOne({roll: 1})
< {
  _id: ObjectId('6b3a7f20d1c94e8b5a60e000'),
  roll: 1,
  name: 'Akash Shukla',
  age: 19,
  marks: 85,
  city: 'Lucknow',
  state: 'UP'
}
```

20. DISTINCT

MongoDB Query:

> *db.students.distinct("city")*

```
> db.students.distinct("city")
< [ 'Delhi', 'Kanpur', 'Lucknow' ]
```